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Determinations of Asteroid Masses Using Mutual Encounters Observed in the Legacy Survey of Space and Time

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Abstract

Using published simulations of the 10 yr Legacy Survey of Space and Time (LSST), we forecast its ability to determine the masses of individual main-belt asteroids (MBAs) through precise astrometry of any pairs of the ≈1.2 million known MBAs undergoing close gravitational encounters during the survey. The uncertainty σ_I on the impulse applied to a tracer asteroid by its deflector is derived from the Fisher matrix of the tracer's astrometric data, including an azimuthal acceleration from the Yarkovsky effect as a free parameter for each tracer. If only LSST observations are available, the mean forecasted σ_I is 7×10^{-6} m s^{-1} for MBAs at apparent magnitude $m_V < 19.5$, degrading ≈10× for $m_V = 23$. The forecasted median uncertainty on the mass of a deflector MBA is $\approx 3 \times 10^{-14} M_\odot$, with a wide range of variation depending on the (random) configurations of the closest encounters. All of these figures improve ≈ fivefold if strong pre-LSST astrometry is available. Use of LSST data will increase the number of MBAs with masses measured to 20% accuracy or better from the present ≈70 to between 200 and 550, depending on quality of pre-LSST data, if all MBAs have mass-to-reflected-light ratios (M/Ls) at the low end of the currently measured bodies. If high-end M/Ls are more common, 580–1100 bodies will attain a signal-to-noise ratio > 5, including nearly all bodies with absolute magnitude $H < 10$ and some as faint as $H = 13$. We forecast that ≈20 new Jupiter Trojans could also have masses measured at <20% accuracy. Tables of the measurable deflector MBAs and their tracers are provided.

Unified Astronomy Thesaurus concepts: Main belt asteroids (2036); Ephemerides (464)

Materials only available in the online version of record: machine-readable tables

1. Introduction

Determination of the masses of asteroids is essential to understanding their composition and history, and for a complete understanding of the dynamics and precision ephemerides of the solar system. Knowledge of asteroid masses has been summarized by A. Fienga et al. (2020), and aggregations of the (often inconsistent) published measures of asteroid masses are maintained at SSODNET[1] (J. Berthier et al. 2023) and SIMDA[2] (M. Kretlow 2020). There are multiple methods to measure a minor planet's mass without assuming a density: high-precision values can usually be inferred for bodies having well-tracked natural satellites or visiting spacecraft. As summarized by A. Fienga et al. (2020), spacecraft tracking data have yielded masses for just four main-belt asteroids (MBAs): (1) Ceres and (4) Vesta, whose summed mass of 1.197×10^{21} kg is roughly half of the total main-belt mass, are known to ≪1% accuracy; plus (21) Lutetia and (253) Mathilde have masses known to a few percent. Observations of MBAs' natural satellites can also yield total masses. According to SIMDA, 17 MBAs have masses derived to ⩽20% accuracy (1σ) by this method as of 2025 April.

A second method is to infer asteroid masses from an ephemeris solution to high-precision astrometry and ranging of the major planets, with the centimeter-accuracy ranging to Mars being of the most power. A. Fienga et al. (2020) used this method to detect masses at a signal-to-noise ratio (S/N) > 3 for 104 MBAs. They obtained subpercent precision (S/N > 100) on Ceres and Vesta, and <10% uncertainty (S/N > 10) on seven other bodies (including Pallas and Hygeia, the next-most-massive bodies). The JPL DE440 ephemeris (R. S. Park et al. 2021) and the EM2017 ephemeris (E. V. Pitjeva & N. P. Pitjev 2018) include ≈300 MBAs as gravitating bodies, but the masses of these bodies are either held fixed to values determined by other means, or do not have uncertainties reported.

The third method for determining asteroid masses, and the one we focus on in this paper, is by fitting to precision astrometric data for pairs of asteroids that undergo a mutual close encounter. Adopting the impulse approximation for these encounters,[3] if the astrometric data can determine the size I of the impulse applied to the tracer body, then we can infer the mass of the deflector body with knowledge of the relative speed v and the separation $\boldsymbol{b}$ of the bodies at the time of closest approach. Approximating the bodies' motions as inertial through the encounter yields the standard formula

$$I = \frac{2GM}{bv}. \tag{1}$$

The mutual-encounter method has been used successfully for roughly 50 yr (H. G. Hertz 1966). J. Baer et al. (2011) reported masses at significance S/N ⩾ 3 for 26 MBAs derived from 82 encounters. Of these, three obtain subpercent accuracy (Ceres,

[1] https://ssp.imcce.fr/webservices/ssodnet/
[2] https://astro.kretlow.de/?SiMDA

[3] Analyses of the data may require going beyond the impulse approximation, but it suffices for forecasting accuracy in nearly all cases.

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Vesta, and (15) Eunomia), and nine more yield $S/N > 10$. W. Zielenbach (2011) fit for masses of 104 MBAs, using a large number of tracer bodies for each, obtaining $S/N > 100$ for Ceres and Vesta, $10 < S/N < 100$ for nine MBAs, and $3 < S/N < 10$ for 22 more asteroids.

The review by J. L. Hilton (2002) reported just 24 known asteroid masses at that time, with some increase in the following decade from the methods described above. SIMDA lists (as of 2025 April) 13 MBAs, beyond those with masses from satellites, with encounter masses yielding <10% accuracy and another ≈40 with accuracy between 10% and 20% (i.e., $5 < S/N < 10$). This gives a total of 73 MBAs with masses measured at $S/N > 5$, according to SIMDA; SSODNET reports 103 MBAs measured at this level of accuracy. The existing data are highly heterogeneous in quality, with very limited information on selection functions or other information that would be needed to draw conclusions about the full population.

Among the Jupiter Trojans (henceforth simply "Trojans"), only three bodies have mass estimates in SSODNET: (617) Patroclus has a 2% measure, (624) Hektor a 17% measure, and (3548) Eurybates a 2% measure, all from binaries. The Lucy spacecraft will fly by Patroclus, Eurybates, and three other Trojans by 2033.

Advances in observational techniques in the 2010s and 2020s should, however, yield a substantial expansion in the number and accuracy of mass determinations from mutual encounters, as well as yielding characterization of nondetections as would be needed for quantitative inferences. First, systematic sky surveys have vastly increased the number of known asteroids, i.e., potential tracer particles, to over 1 million. Second, the Gaia Data Release 3 (Gaia Collaboration et al. 2018) included milliarcsecond-precision positions for >23 million epochs of >150,000 solar system bodies with apparent magnitudes $G < 21$ over 34 months of operation, extended to 66 months in Gaia Collaboration et al. (2023). While the number of new asteroid masses reported from Gaia DR3 has been modest to date (F. Li et al. 2023; Z. Murray 2023), the Gaia spacecraft operated for >10 yr, and later data releases plus further analyses should yield a larger number of mass detections as well as better mass accuracy per encounter. Perhaps more importantly, Gaia has defined a stellar reference frame that allows ground-based observatories to place asteroid positions onto an inertial coordinate system with systematic errors well below 1 mas. The Legacy Survey of Space and Time (LSST), to be conducted with the Vera C. Rubin Observatory from 2025–2035, will detect and track millions of MBAs, with accuracy limited by a combination of (magnitude-dependent) photon noise and ≈2 mas uncertainty from refraction by atmospheric turbulence (W. F. Fortino et al. 2021). Each source will be detected 200–600 astrometrically useful times over the course of the LSST. The per-observation astrometric uncertainties for LSST are smaller than those for Gaia at the faint limit of the latter, and LSST's observations will extend ≈3 mag deeper, increasing the pool of potential tracers by ≈100×. As noted in the LSST Science Book (LSST Science Collaboration et al. 2009) "At present only a few dozen asteroids have mass estimates based on perturbations, but LSST will produce astrometry that is both prolific and precise, at the same time that it dramatically expands the pool of potential test particles. LSST data should allow the estimation of the mass of several hundred or so main-belt asteroids with an uncertainty of ∼30% or less."

In this paper, with LSST now on the verge of operation, we quantify this prediction more carefully by finding all encountering pairs among all MBAs and Trojan asteroids that should yield deflections of ≈1 mas or greater over the 10 yr observing period of LSST, and using a simulation of LSST's observing schedule and performance to forecast the uncertainties in the masses of the deflectors in these encounters. While the exact sequence of observations and astrometric errors of LSST are both, of course, not yet known, we are at this time able to make accurate estimates of the future performance of the survey, to know what fractions of the asteroids of what masses will be measured to useful significance using presently known bodies as tracers. For purposes of inferring an individual asteroid's bulk density, we will consider a 1σ error of <20% on the mass, or $S/N > 5$, as the point where the mass information starts to become "useful." Once we reach $S/N \approx 10$, additional precision on the mass will not generally lead to greater compositional knowledge, since uncertainties in the albedo/diameter/volume of the body will dominate the density uncertainty. For dynamical purposes, however, it is the typical uncertainty σ_M of the MBA' masses (rather than the S/N) that determines errors in ephemerides.

The forecast for knowledge of asteroid masses arising from mutual encounters during LSST's 10 yr lifetime still depends fairly strongly on the quality of pre-LSST observations of the tracer body in each event. We will bound the possibilities by making one "LSST-only" forecast, assuming the pre-LSST data offer no useful information, and one "Initial-known" forecast, assuming that pre-LSST data perfectly determine the tracers' state vectors ($\boldsymbol{x}_0$, $\boldsymbol{v}_0$) at the start of LSST. In both scenarios, the transverse Yarkovsky acceleration coefficient A_2 for each tracer will be considered an additional free parameter in the solution for deflector masses. Note that even when assuming strong pre-LSST data, we are confining our forecasts to impulses that occur during LSST. Additional knowledge could be gained from impulses that occur between earlier observations and the start of LSST.

We are not claiming to be exhaustive in our inventory of bodies whose masses LSST will determine, since the precise list will depend upon the exact realization of the survey, and the results may also be influenced by tracer asteroids that are not yet discovered. We also have not attempted to evaluate the accuracy of pre-LSST observations for each tracer body, but it is clear that this will be a critical activity for getting the most asteroid-mass knowledge out of LSST.

2. Identifying Encounters

2.1. Asteroid Properties

The search for asteroid–asteroid encounters with potentially detectable effects on LSST astrometry begins with the `MPCORB.DAT` file containing H values and osculating heliocentric orbital elements for all asteroids known to the Minor Planet Center (MPC).[4] We retain from this file the 1.26 million objects with observations at multiple oppositions, semimajor axes $a < 6$ au, and MPC uncertainty indicator $U \leqslant 5$. We also exclude Ceres, Vesta, and Pallas from the list since they will generate very large numbers of encounters above our impulse threshold that are not of interest—the first two already have their masses very well determined by

[4] Downloaded 2025 January 6 from the MPC data page: https://www.minorplanetcenter.net/data.

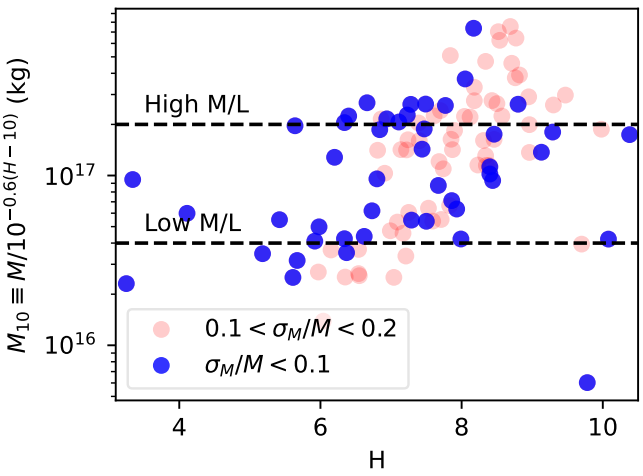


Figure 1. The mass-to-light ratio M_{10} defined in Equation (2) is plotted vs. absolute magnitude H for all MBAs in the SSODNET database (J. Berthier et al. 2023) with mass measurements accurate to 20% or better. For our forecasts, we will consider the two values of M_{10} marked by the horizontal lines labeled "low M/L" and "high M/L."

spacecraft, and Pallas is already approaching percent-level accuracy from previous data on mutual encounters.

We will require an estimate of the mass M of each asteroid at three points in our forecast:

1. Determining the largest impact parameter $b_{\max}$ that might generate an impulse of use in constraining the mass of a given deflector.
2. Placing a prior on the amplitude A_2 of the acceleration of the tracer caused by the Yarkovsky effect.
3. Determining the S/N M/σ_M for the mass determination of the deflector after we have derived the uncertainty σ_M achievable from data on its tracers.

The mass M of an MBA with absolute magnitude H is of course highly uncertain—determining M values is the point of the experiment we are forecasting! Our forecasts of the attained uncertainty σ_M depend very weakly on the $M(H)$ choice—entering only through the Yarkovsky prior. The number of MBAs attaining a given S/N ratio depends at first order, however, on the choice of $M(H)$. We will thus need to examine a fairly broad range of possibilities for $M(H)$.

For an object with spherical shape, geometric albedo q, and density ρ, the standard formula[5] for radius versus H and q yields

$$M = M_{10} 10^{-0.6(H-10)}, \tag{2}$$

$$M_{10} = 3.6\left(\frac{q}{0.1}\right)^{-3/2}\left(\frac{\rho}{1000\ \mathrm{kg\ m^{-3}}}\right) \times 10^{16}\ \mathrm{kg}. \tag{3}$$

In the first line, we define M_{10} as the mass of a body with $H = 10$. Because more bodies have known masses than have both density and albedo known, we will choose candidate values of the "mass-to-light ratio" M_{10} directly rather than choosing values of ρ and q to insert into Equation (3). Figure 1 plots the value of M_{10} versus H for all MBAs with masses known to 20% accuracy or better according to the SSODNET database. The derived M_{10} values span a factor of >20 even ignoring potential outliers. The known M_{10} values can very crudely be described as dividing into two groups, each lying within a factor of ≈2 of one of the values marked by horizontal lines in the figure:

1. The "low-mass-to-reflected-light-ratio, M/L" value $M_{10} = 4 \times 10^{16}\,\mathrm{kg} = 2 \times 10^{-14} M_\odot$, as could arise from a spherical body with $q = 0.15$ and $\rho = 2000\,\mathrm{kg\,m^{-3}}$.
2. The "high-M/L" value $M_{10} = 2 \times 10^{15}\,\mathrm{kg} = 10^{-13} M_\odot$, as could arise from a spherical body with $q = 0.04$ and $\rho = 1400\,\mathrm{kg\,m^{-3}}$.

The $H > 8$ bodies seem to favor the high-M/L value, but we should remember that this diagram is subject to many potential systematic errors and selection effects, particularly those that favor reporting of values only for higher-mass bodies at a given H. When identifying potentially interesting encounters, adopting the high-M/L scenario will yield a superset of the encounters of interest in the low-M/L scenario. When we forecast the number of deflectors attaining a given S/N on mass, we will examine results of both scenarios. The threshold for measurement of MBA mass at a given S/N moves faintward in H by 1.2 mag, for the high-M/L value of M_{10} compared to the low-M/L value.

2.2. Encounter Search

The osculating heliocentric orbital elements of the asteroids are converted to barycentric state vectors in the ICRS coordinate frame, and integrated forward in time to the start date of our simulated LSST survey using a simple leapfrog integrator with time step of $\Delta t = 1$ day. The integrator assumes Newtonian dynamics, with accelerations derived from the positions of the Sun and eight planets. Masses and positions for the gravitating bodies are taken from the JPL DE440 ephemeris.[6] While measurement of asteroid masses from real data with milliarcsecond accuracy will require more sophisticated integrators, incorporating relativistic effects and perturbations from some of the larger asteroids, the forecasting of inferences does not require great accuracy. The ephemeris accuracy must only be sufficient to successfully predict (1) the impact parameters and relative velocities of encounters to ≲10% accuracy, and (2) the derivatives of the observed position with respect to initial state vector and the impulse velocity, to similar accuracy. Because the closest encounters have $b \approx 10^{-5}$ au, integrator accuracy of 10^{-6} au or ≈150 km is more than adequate.

We wish to find all encounters during the 10 yr survey that will generate a statistically significant shift in the tracer's LSST measurements. Based on estimated accuracy of ≳1 mas for LSST astrometry of tracers, we make a rough estimate that encounters generating impulses of $I < I_{\min} = 10^{-6}\,\mathrm{m\,s^{-1}}$ will never be detectable at ≳3σ in the LSST astrometry of a tracer —this is confirmed in Section 4. We hence will seek only those encounters capable of generating $I > I_{\min}$ if the deflector has the high-M/L value.

From Equation (1), we can derive a maximum impact parameter $b_{\max}$ between a deflector of mass M and a tracer moving at relative velocity $v_{\min}$ that will result in an impulse above $I_{\min}$. Initial investigation of the distribution of relative velocities in encounters shows <100 out of 4 million tabulated

[5] Exempli gratia, this table: https://www.minorplanetcenter.net/iau/Sizes.html.

[6] Downloaded from this JPL site: https://naif.jpl.nasa.gov/pub/naif/generic_kernels/spk/planets/.

encounters have relative velocity below $v_{\rm min} = 100\ {\rm m\ s^{-1}}$, so we are not losing significant information by setting $b_{\rm max}$ with this $v_{\rm min}$ value. Slower encounters at $b < b_{\rm max}$ will still be included in our forecast. Any super-slow encounters that we miss because they are at $b > b_{\rm max}$ will have encounter times that will no longer be a small fraction of the orbital period, so these would need to be modeled in more detail than the impulse approximation. We obtain

$$b_{\rm max}(H) = \frac{2GM}{I_{\rm min} v_{\rm min}}. \quad (4)$$

Inserting our high-M/L value of M_{10} into Equation (4) yields

$$b_{\rm max}(H) = 1.6 \times 10^{-0.6(H-10)}\ {\rm au}. \quad (5)$$

Using the low-M/L M_{10} would yield a lower $b_{\rm max}$. We will limit our search to $b < 0.1$ au, however, since the numerical results show that it is rare for the most informative encounter for a given deflector to be an event with $b > 0.1$ au.

The asteroids are split into groups spanning intervals of 1 mag in H, and $b_{\rm max}(H)$ is evaluated at the brightest H value for each group. We next enter a loop to identify encounters occurring during the time period 2025 May 1 through 2035 May 1 of the simulated LSST.[7] The loop begins with t at the start date of the simulated survey. Each iteration contains these steps:

1. If t is after the survey end date, end the loop.
2. Build a k-d tree of the spatial positions $x(t)$ of the asteroids in each bin of H.
3. Use the (fast) k-d tree algorithms for finding close pairs between each combination of deflector H bin and tracer H bin. The matching radius is set to

$$r_{\rm max} = \sqrt{\min[0.1\ {\rm au},\ b_{\rm max}(H_{\rm deflector})]^2 + (v_{\rm max} t_{\rm step}/2)^2}\ , \quad (6)$$

   where $v_{\rm max} = 3 \times 10^4\ {\rm m\ s^{-1}}$ is the largest expected relative velocity for an interesting encounter, and $t_{\rm step} = 2$ days is the time step for the loop. This expression is the relative separation between deflector and tracer asteroid at time t, assuming that they reach the closest approach at time $t_{\rm imp}$ such that $|t - t_{\rm imp}| \leqslant t_{\rm step}/2$.
4. For each (deflector,tracer) pair found within $r_{\rm max}$, we find the time $t_{\rm imp}$, and vector displacement $\boldsymbol{b}$ of closest approach, and relative velocity v, assuming inertial motion with the positions $x(t)$ and velocities $v(t)$ of the two asteroids.
5. If $|t_{\rm imp} - t| > 0.55 t_{\rm step}$, i.e., the closest approach is not within this time step, we discard the pair. It should be found again and retained at an earlier or later time step.
6. We calculate the expected impulse I generated by the encounter, using the mass derived from the deflector's H for the high-M/L case. If $I < I_{\rm min}$, we discard the pair.
7. We use the leapfrog integrator to advance the state vectors and the time t by $t_{\rm step} = 2$ days.

Only those encounters occurring between 1 and 9 yr after survey start are retained, on the assumption that the others have insufficient data before or after the deflection to allow for a useful measure of the impulse. If a given deflector-tracer pair register multiple encounters during this time period, we retain only the one producing the highest impulse, simplifying our analysis at a small cost in accuracy of the forecast.

The results of the search for encounters are illustrated in Figure 2, in which we consider only those encounters that would generate an impulse $I > 2 \times 10^{-6}\ {\rm m\ s^{-1}}$ assuming the high-M/L value of M_{10}. All of the 725 MBAs with $H < 10$ have potentially measurable encounters with a known tracer, even if the lower-M/L value of M_{10} is assumed. Indeed, most $H < 10$ MBAs perturb many tracers, into the thousands, even though we have truncated our search to $b < 0.1$ au. In further analyses, we will only examine the 20 encounters with any given deflector that generate the highest impulses, since we expect that most of the information on its mass will be provided by the tracers receiving these 20 largest impulses.

At $H \gtrsim 11$, the probability of a deflector encountering a tracer at small enough b and $v_{\rm imp}$ to generate an impulse $I > 2 \times 10^{-5}\ {\rm m\ s^{-1}}$ becomes less than unity—with this transition happening at brighter H if we assume the low-M/L value of M_{10}. The cross section for a measurable impact shrinks as $b_{\rm max}^2 \propto M^2 \propto 10^{-1.2H}$, which is steeper than the growth in the number of the potential MBA deflectors, $dN/dH \propto 10^{0.5H}$. This means that the number of deflector MBAs generating detectable encounters drops fairly quickly when H increases in this incomplete-census regime.

If we now restrict our analysis to the 20 largest-impulse encounters for any MBA deflector having > 20 encounters on the candidate list, we are left with ≈53,000 encounters involving ≈7400 unique deflectors. Roughly 4% of the input list of MBAs is tracers in one or more cataloged deflections. The number of MBAs receiving impulses above $I_{\rm min}$ from $\geqslant 2$ distinct deflectors is ≈3% of all deflected MBAs, consistent with Poisson statistics for encounters between deflectors and tracers. In further analyses, we will take the shortcut of analyzing each encounter of a given tracer independently, neglecting the covariances between mass estimates from these occasional multiple-encounter tracers.

3. Dynamical Model

The next step is to quantify the observational signatures of deviations from the nominal MPC orbit for the tracers of each of the 53,000 encounters. Free parameters of the tracer orbit are the six elements of the state vector at the start of LSST, plus the applied impulse I. We need also to consider nongravitational forces on the tracer body if (1) the astrometric deviations they cause, or similarly their integrated impulse on the tracer, are larger than ≈1/10 of the effects the gravitational encounter, *and* (2) the time signature of the effect resembles that of an impulse, such that a fit to the data would find the nongravitational force covariant with the size of I.

Radiation pressure satisfies criterion (1) for smaller tracers. The acceleration $a_{\rm rad}$ from incident radiation pressure on a spherical body of radius R and density ρ at distance r from the Sun is

$$a_{\rm rad} = \frac{\pi R^2 L_\odot}{4\pi r^2 M c} = \frac{R^2 L_\odot}{4Mc}\frac{1}{r^2} \quad (7)$$

[7] The real LSST is currently scheduled to begin later in 2025; this will change the details of individual asteroids' mass measures, but not the statistics of the full population.

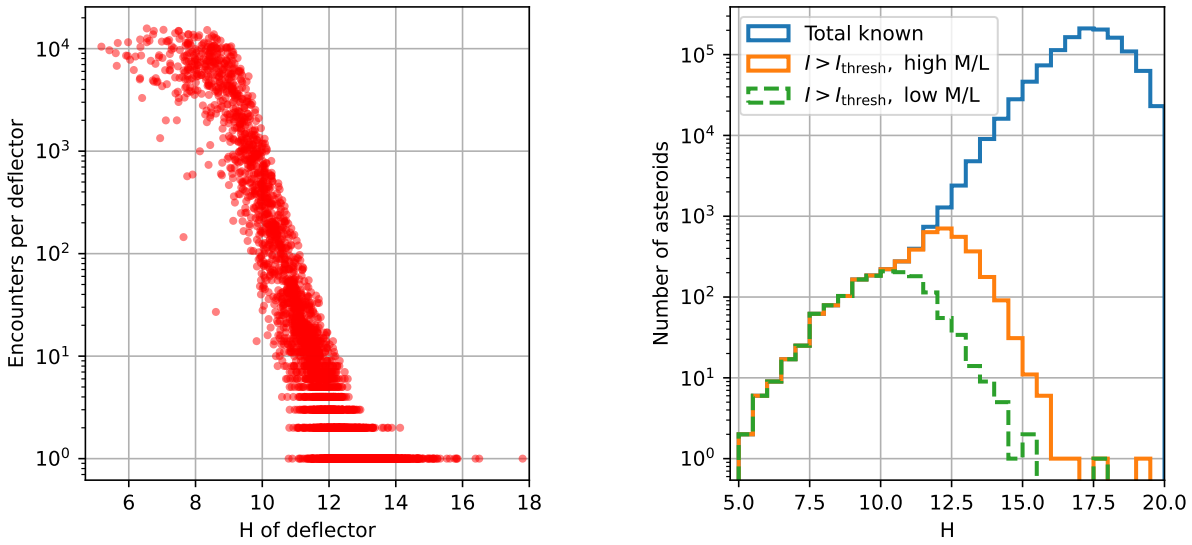


Figure 2. The results of the search for useful deflector-tracer encounters among MBAs in a simulated LSST are plotted vs. the absolute magnitude H of the deflector. The left plot gives the number of encounters producing impulses $I > 2 \times 10^{-6}$ m s^{-1} for each unique deflector. This number is well above unity for $H \lesssim 10.5$, if we assume the high-M/L value of M_{10}. This is despite the fact that we retain only encounters with impact parameter $b < 0.1$ au. The right plot's blue histogram shows, in bins of 0.5 mag in H, the total number of known MBAs at $a < 5$ au. In orange (dashed green) is the histogram of those MBAs that perturb at least one tracer with an impulse $>2 \times 10^{-6}$ m s^{-1}, assuming the high (low) M/L value for M_{10}. As expected from the left plot, all of the 725 MBAs at $H \lesssim 10$ generate an impulse above this threshold on a known tracer during the LSST. Among the fainter/smaller bodies at $H > 10$, not all generate an above-threshold impulse, but some fraction of them is "lucky" in that they encounter a tracer with small enough impact parameter and velocity. In this way, an incomplete but random selection of MBAs at $H > 10$ are candidates for mass measurement, outnumbering the complete sample at $H < 10$. Assuming the lower-M/L value for all MBAs shifts the completeness curve to brighter H, as expected.

$$\approx 1.5 \times 10^{-14} \times 10^{0.2(H-10)} \text{ m s}^{-2} \times \left(\frac{1 \text{ au}}{r}\right)^2 \left(\frac{0.1}{q}\right) \left(\frac{4 \times 10^{-16} \text{ kg}}{M_{10}}\right). \quad (8)$$

The value in the last line adopts as nominal values the low-M/L value of M_{10}, and a geometric albedo $q = 0.1$, which is on the low end of what a low-M/L asteroid might have. This yields a larger, i.e., more conservative, estimate of the potential size of nongravitational acceleration. Applying this acceleration over 10 yr yields a total velocity shift that is, for many tracers, comparable to or larger than the typical forecasted uncertainties in gravitational impulses.

The incident radiation pressure does not, however, satisfy criterion (2), since a $1/r^2$ radial force does not have observable consequences similar to those of an instantaneous impulse. The orbit is altered only by being at very slightly larger distance for a given period, at a level that is undetectable in LSST astrometry. Similarly, the reflex pressure from reflected sunlight should yield a nearly radial $1/r^2$ force and can be ignored here. More intrusive, however, is the net reaction force from anisotropic thermal reradiation of absorbed solar flux, i.e., the Yarkovsky effect. The component along the radius vector $\hat{\boldsymbol{r}}$ once again is not covariant with a gravitational impulse, but the component a_Y directed in the azimuthal direction (perpendicular to $\hat{\boldsymbol{r}}$, in the orbital plane) must be considered.

The component of an impulse that is parallel to the tracer's velocity (nominally azimuthal) generates the largest observable signal because it changes the tracer's energy and, hence, its semimajor axis a, and its period P. This causes a perturbation in the tracer's ecliptic longitude that grows linearly with time after the impulse. The azimuthal Yarkovsky force changes a at a constant rate, leading to an azimuthal perturbation that grows quadratically with time. This quadratic Yarkovsky signal is covariant with the zero-then-linear signal from an encounter, degrading the accuracy of the inference on I. Longer periods of observation make it easier to disentangle these effects.

We therefore include an azimuthal Yarkovsky acceleration $a_Y = A_2(1 \text{ au}/r)^2$ in the dynamical model of the tracer. We can in general write

$$a_Y = \gamma g_Y a_{\rm rad}, \quad (9)$$

where γ is the fraction of incident radiation absorbed at the surface, and g_Y is a factor giving the average azimuthal component of the angular radiation pattern. For a spherical body with a homogeneous, Lambertian surface, the maximum attainable value of g_Y is $\approx$0.1, for the diurnal Yarkovsky effect (e.g., S. Tremaine 2023). The maximum is attained when the spin axis of the asteroid has obliquity $\epsilon = 0$ or π, i.e., perpendicular to the orbital plane, and the thermal/rotation properties reach an optimum value. The diurnal effect scales as $\cos\epsilon$, so can be either a prograde or a retrograde force, just as a gravitational impulse can be. An isotropic distribution of spin axes would yield an rms variation of a_Y equal to $1/\sqrt{3}$ of its maximal value, and variations in thermal properties from the optimum would lower this further.

There are other subtleties to the Yarkovsky force—a seasonal effect, binary effects, and results of non-Lambertian surfaces (see review by D. Vokrouhlický et al. 2015)—but these would typically enter at lower amplitude and/or with temporal patterns less resembling those of an impulse. We will therefore limit our Yarkovsky treatment to having a single free parameter A_2 for each tracer, and will place a prior probability distribution on A_2 that is Gaussian with a standard deviation implied by Equations (8) and (9) with the H value of the tracer, $q = 0.1$, our low-M/L M_{10} value, and a value of $\gamma g_Y = 0.05$. The resultant typical integrated impulses from the Yarkovsky effect are therefore 20× smaller than that of the full incident

radiation pressure $I_{\rm rad}$ estimated above—but still have significant impact on forecasted mass measurements, as we shall quantify below.

We then conduct another leapfrog integration of each tracer asteroid's path during the time period of the simulated LSST. In this integration, we consider the tracer subject to the Yarkovsky force in addition to the gravity of the Sun and eight planets. We want the partial derivatives of the tracer's R.A. and decl. $(\alpha,\ \delta)$ with respect to the parameters $\boldsymbol{q} = \{x_0, y_0, z_0, \dot{x}_0, \dot{y}_0, \dot{z}_0, I, A_2\}$, as function of time t of observation. We obtain this by integrating eight test particles that have had the nominal MPC orbit perturbed by small deviations to each parameter. This yields a time-dependent 2×8 derivative matrix $\partial[\alpha(t), \delta(t)]/\partial\boldsymbol{q}$. The apparent V-band magnitude of the tracer asteroid is saved as well, for purposes of forecasting the observational errors. We ignore phase corrections for this work.

4. Forecasting LSST Observations and Impulse Uncertainties

With a list of interesting encounters in hand, we must now quantify the ability of LSST to measure the positions of the relevant tracer asteroids. We use the table of simulated LSST exposures given in `baseline_v4.0_10yrs.db`, produced by the Rubin Observatory Survey Cadence Optimization Committee group.[8] The simulations emulate a sequence of all exposures taken for the main "Wide Fast Deep" survey, and the circumstances of each, including time, sky coordinates, filter, sky conditions, and point-spread function (PSF) size. The methods for finding the exposures containing a given asteroid, and estimating the rms measurement error σ_θ on each axis of its sky position, are the same as used by D. C. H. Gomes et al. (2023). In brief: for each tracer MBA, we find all LSST exposures that would contain its image. The astrometric uncertainty that would be attained on the MBA has two components. The first is a photon-noise error, which can be calculated using the estimated quantum efficiency of the observatory along with the MBA's apparent magnitude, and the circumstances of the (simulated) exposure: filter band, sky background, cloud cover, airmass, and width of the PSF. The second component is a fixed 2 mas, added in quadrature to the photon-noise error to account for LSST astrometric calibration uncertainties and the stochastic displacements from atmospheric turbulence, after correction using the techniques developed by W. F. Fortino et al. (2021). The result is an estimate σ_{ij} of the rms uncertainty along each of the α and δ axes for the position of tracer i in exposure j.

The uncertainty in the impulse I applied to tracer i is estimated using the Fisher information matrix F_i for its observations, defined as

$$(\mathsf{F}_i)_{uv} = \sum_j \frac{1}{\sigma_{ij}^2}\left[\frac{\partial\alpha(t_j)}{\partial q_u}\frac{\partial\alpha(t_j)}{\partial q_v}\cos^2\delta(t) + \frac{\partial\delta(t_j)}{\partial q_u}\frac{\partial\delta(t_j)}{\partial q_v}\right], \quad (10)$$

where u, v run over the eight elements of the parameter vector $\boldsymbol{q}$ for this MBA, and the sum runs over LSST observations of tracer i.

The Cramer–Rao theorem then provides a lower bound on the uncertainty σ_I of the inference of the impulse on a given tracer:

$$\begin{aligned}
\sigma_I^2 &= [\mathsf{F}^{-1}]_{II} \quad \text{marginalizing over } \boldsymbol{x}_0,\ \boldsymbol{v}_0,\ A_2,\\
\sigma_I^2 &= [\mathsf{F}_{A_2 I}^{-1}]_{II} \quad \boldsymbol{x}_0,\ \boldsymbol{v}_0 \text{ known, marginalizing } A_2,\\
\sigma_I^2 &= (\mathsf{F}_{II})^{-1} \quad \boldsymbol{x}_0,\ \boldsymbol{v}_0,\ A_2 \text{ known.}
\end{aligned} \quad (11)$$

The notation $\mathsf{F}_{A_2 I}$ indicates truncation of the full Fisher matrix to the 2×2 submatrix for parameters A_2 and I. The first case, which we label "LSST only," corresponds to using only LSST data to estimate the impulse I_i simultaneously with the initial state vector $(\boldsymbol{x}_0,\ \boldsymbol{v}_0)$ and Yarkovsky strength A_2. The middle "Initial-known" case is when pre-LSST data give $\boldsymbol{x}_0$ and $\boldsymbol{v}_0$ to high accuracy, but A_2 remains to be fit from the combined data. The third "Ideal" case is the limit where $\boldsymbol{x}_0$, $\boldsymbol{v}_0$, and A_2 are known precisely from pre-LSST observations—this case is unrealistic since there are no published detections to date of A_2 for individual MBAs. But it will be of interest to see how much the Yarkovsky effect is degrading mass accuracy.

The σ_I determined from simulated LSST data of a tracer is independent of the amplitude of the impulse I, meaning it is also independent of the mass of the deflector, the impact parameter $|\boldsymbol{b}|$, and the relative velocity v. This means that our 53,000 tracers are effectively a random sample of the σ_I available from all of the known MBAs. The forecasted σ_I will depend primarily on the apparent magnitude m of the tracer, which determines the LSST measurement errors; also rather heavily on the date of the impulse relative to the start and end of the survey; and on the direction of the impulse. There is also stochasticity in the number and quality of a tracer's observations by LSST.

Figure 3 plots several properties of the σ_I values for the tracer population. Of note, the information-content mean σ_I, defined as $\langle\sigma_I^{-2}\rangle^{-1/2}$, does degrade for fainter tracers, roughly as $\sigma_I \propto 1/\text{flux}$ (upper row of the Figure). The middle row shows that this leads the distribution of useful tracers to peak at apparent magnitudes 21–22, roughly where the raw MBA counts peak, suggesting that discoveries of a large number of fainter MBAs by LSST could increase the number of useful encounters and improve over the results we derive. We also see that there is substantial degeneracy between I and the initial state vector in the orbit fits, because fixing the latter, as we move from left to central column, lowers the uncertainty on the former by $\approx 5\times$. Advance knowledge of the Yarkovsky strength, indicated by moving from the center to right columns, improves σ_I by another factor of ≈ 2, for the fainter tracers, but very little for the brightest ones.

The bottom row of Figure 3 also confirms that our initial choice to investigate only those mutual events causing $I > 10^{-6}\,\mathrm{m\,s^{-1}}$ has not excluded any significant number of encounters that could have been useful. In the left panel, we see that none of the tracers are capable of detecting this chosen $I_{\rm min}$ when only LSST data are used. The center and right panels shows that even with strong pre-LSST orbital knowledge, only a tiny fraction of tracers have $\sigma_I < I_{\rm min}$, and none could detect $I_{\rm min}$ with $\mathrm{S/N} > 3$. The lowest σ_I values are, as expected, attained by brighter tracer MBAs.

5. Resulting Deflector Mass Uncertainties

For each deflector-tracer encounter, we have derived σ_I for the tracer's LSST astrometry. This can be propagated into an uncertainty $\sigma_M = (bv/2G)\sigma_I$ using the circumstances of the

[8] Simulation files are linked and documented at the Committee's site: https://survey-strategy.lsst.io/baseline/index.html.

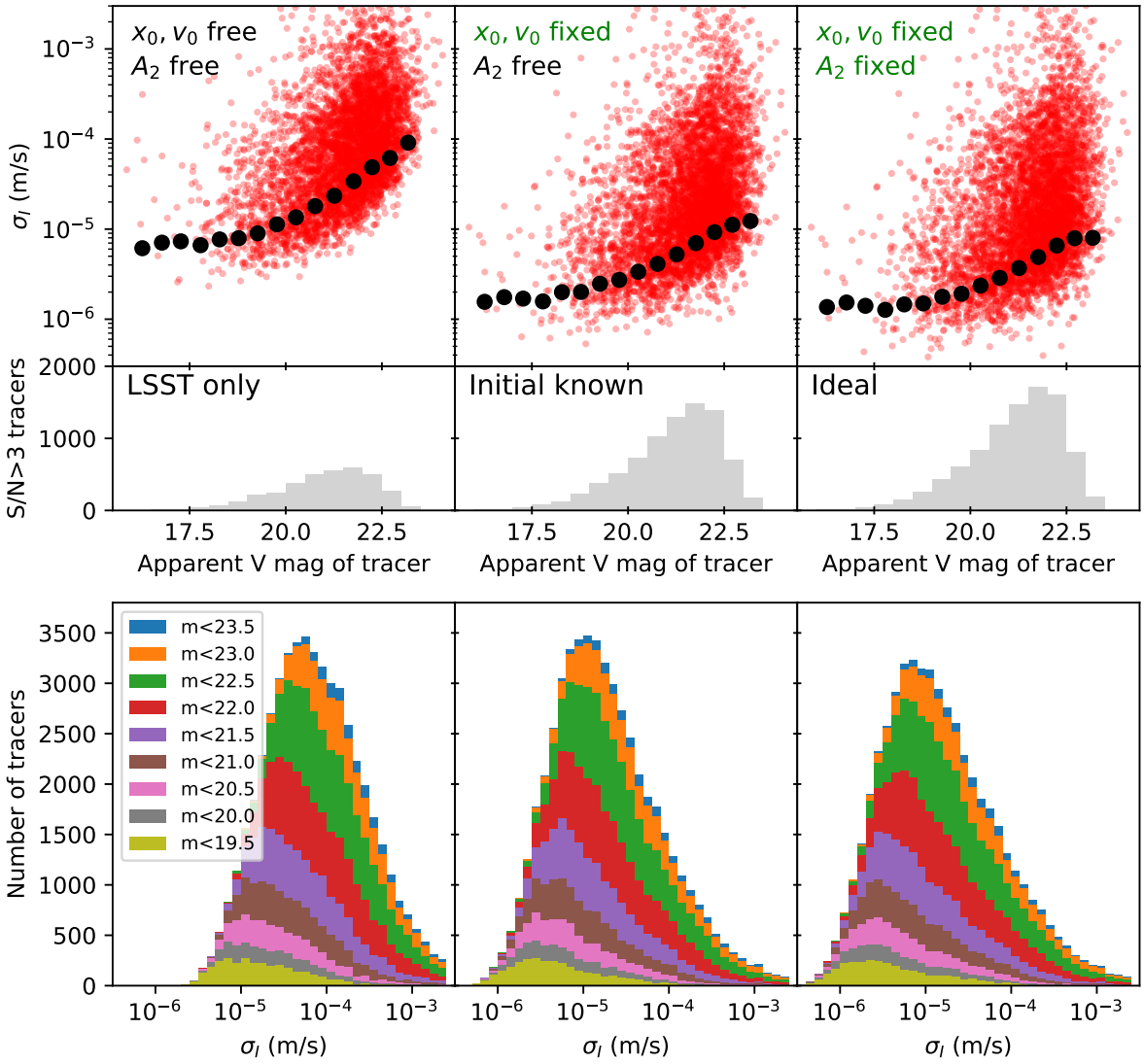


Figure 3. Each plot gives information about the uncertainty σ_I on the applied impulse that is forecasted for observations of each individual tracer MBA of a mutual encounter. The columns assume progressively more information available from pre-LSST data about the tracers' initial state vectors and Yarkovsky accelerations when going from left to right, as denoted in the text atop each column. The top row of plots is a scatter diagram (red dots) of σ_I vs. the apparent magnitude of the tracer; the black dots trace the mean vs. apparent magnitude. The middle panels give the distribution of the apparent magnitudes of those tracers that determine their deflectors' masses with S/N > 3, i.e., the distribution of apparent magnitudes of the useful tracers. The bottom panels plot the distributions of σ_I for tracers of various magnitudes.

encounter. Because the more-massive deflectors are involved in multiple deflections with $I > I_{\rm min}$, we examine results that combine the mass measures of all impulses, up to 20, from deflector d:

$$\sigma_M = \left(\sum_i \sigma_{M,i}^{-2} \right)^{-1/2} . \quad (12)$$

Combining information from the 20 most informative tracers improves σ_M by a modest factor ($\approx$2) over using the single best tracer, and in further analysis, we will assume this combination is done. Gains from >20 tracers per deflector are minimal.

Figure 4 plots the distributions of σ_M for the MBA deflectors. The distribution of σ_M should be independent of the deflector mass; however, our imposition of a minimum on the applied impulse means that only for the $H < 10$ deflectors are we locating most of the 20 highest-I encounters with tracers in our analysis. Therefore, we plot the distributions of σ_M only for this more-massive subsample of deflector MBAs, as a better representation of what σ_M is attainable. The median deflector has $\sigma_M \approx 10^{-13.5} M_\odot$ in the LSST-only scenario, while the "Initial-known" case of very strong pre-LSST observations has median $\sigma_M \approx 10^{-14.1} M_\odot$, roughly 4× better. The "Ideal" case suggests the median σ_M would decrease another ≈1.4× if we did not have to consider uncertainty in the Yarkovsky coefficient A_2. The median σ_M for the LSST-only case is a 5σ detection of the mass of an $H = 9.6$ asteroid in the high-M/L scenario, $H = 8.5$ in the low-M/L scenario. At the other limit of strong pre-LSST information, the median sensitivity is a 5σ detection at $H = 10.7$ (9.5) for high-M/L (low-M/L) MBAs. In every case, there are tails in the derived σ_M that extend an order of magnitude in each direction from the median, corresponding to raising or lowering the H of 5σ detections by 1.67 mag.

Figure 5 plots our forecasts for how many MBAs will have their masses determined to S/N > 5 or S/N > 10 through their

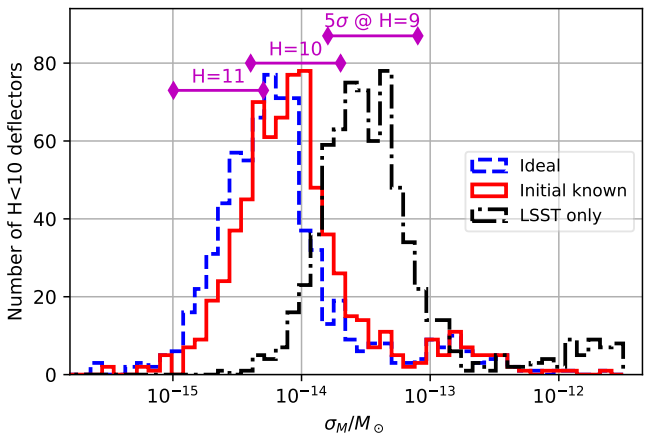


Figure 4. The distribution of forecasted σ_M, the uncertainty on the mass of the deflector, is plotted for all of the 725 MBA deflectors with $H < 10$. The dashed–dotted black histogram is the "LSST-only" case, the solid red histogram assumes that pre-LSST data strongly constrain the initial state vector of the tracer, and the dashed blue histogram shows what happens if the Yarkovsky coefficient A_2 is also known precisely. Along the top are marked the ranges of σ_M needed to obtain a 5σ detection of the masses of MBAs at different H values, assuming our low-to-high range of possible masses at a given H.

deflections of other MBAs. If the MBAs have uniformly low M/L, then we expect between 200 and 550 MBAs to obtain S/N > 5 mass measurements, as we proceed from zero pre-LSST information to very strong pre-LSST constraints. This would be a three- to eight-fold increase in the number of masses currently tabulated to this accuracy in SIMDA. If MBAs have our adopted high-M/L value, this increases to 580–1100 objects with 5σ measures, roughly an order-of-magnitude gain over current knowledge.

Counting only MBAs forecasted to obtain $>10\sigma$ detections (10% mass measures): at low M/L, we expect 130–370, and 410–820 if they have high M/L. These are five- to 40-fold increases over the 24 MBAs with this accuracy in SIMDA.

Among the Trojans, 22 bodies (including Hektor and Patroclus) are forecast to obtain S/N > 5 in the case of strong pre-LSST knowledge of the tracers and high-M/L ratio, seven of them reaching S/N > 10. None of these attain S/N > 5 in the LSST-only case, and (1172) Aeneas is the only body that can reach S/N = 5 if they all have the low M/L value. The 22 bodies have $7.3 < H < 10$, and knowing their masses at this level would be of great help in understanding the origin and diversity of the Trojan population.

Full information on all of the forecasted encounters is given in Table 1, namely the identities of the deflector and tracer asteroids, the time and circumstances of the impulse, the nominal size of impulse given the H value of the deflector, and the forecasted σ_I of the tracer in the three scenarios of pre-LSST knowledge. Then, in Table 2, we combine the information from all tracers of a given deflector to give a forecast of the attainable σ_M on each one.

6. Discussion

The conclusions of our forecast of LSST constraints on mutual encounters of asteroids can be summarized in a few different domains. In terms of ability to measure an impulse I occurring sometime during LSST years 2–9, the typical information from observations of an MBA with apparent V-band magnitude $m < 19.5$ measures the impulse to $\sigma_I \approx 7 \times 10^{-6}\,\mathrm{m\,s^{-1}}$ if we use only data from LSST to solve simultaneously for the initial state vector of the tracer, its nongravitational acceleration A_2, and the applied impulse I. If pre-LSST observations constrain the initial state strongly, this improves to $\approx 2 \times 10^{-6}\,\mathrm{m\,s^{-1}}$. These numbers degrade for fainter tracer MBAs, which have higher observational astrometric uncertainties and larger Yarkovsky accelerations, becoming $\approx 10\times$ worse at $m = 23$ mag.

In terms of the uncertainty σ_M on the mass of a deflector that is attained by LSST astrometry of the $\leqslant 20$ tracer MBAs on which it imparts the largest impulses, we find a median value $\sigma_M = 10^{-13.5} M_\odot$ using only LSST data, improving to $10^{-14.1} M_\odot$ as pre-LSST data become capable of constraining the initial state vector. The value of σ_M for a given deflector varies, however, by more than a factor of 10, depending upon the "luck" of a given deflector MBA in having encounters at small distances b and relative velocities v with brighter tracers.

In forecasting the number of MBAs with masses measured to S/N > 5, we need to adopt an assumption for the (currently unknown) mass-to-light ratio distribution of MBAs. If we assume a low nominal M/L ratio, we forecast between 200 and 550 MBA S/N > 5 measurements, depending on the quality of pre-LSST observations. With a high nominal M/L ratio, this grows to 580–1100 MBAs, an order-of-magnitude increase over present knowledge. We find that 60%–75% of the S/N > 5 determinations will attain S/N > 10. Masses uncertainties below 1% are forecasted for 71 (232) MBAs (in addition to the already-known Ceres, Pallas, and Vesta) assuming low (high) M/L ratios for them, including some at $H > 9$. These values assume strong pre-LSST information.

For any choice of S/N threshold, pre-LSST data quality, and M/L ratio, the MBAs with successful mass determinations follow a common pattern. Up to some H (somewhere between 7.5 and 10), corresponding to masses several times the typical σ_M, the mass determinations are nearly complete. But only $\approx 1/2$ of the MBAs with successful mass measures are brighter than this completeness level; the other half are fainter, less-massive MBAs that have fortunately close encounters with one or more known tracer MBAs, yielding σ_M values more precise than the median. Thus, we expect to gain useful mass information on an essentially random subsample of MBAs down to $\approx 100\times$ lower mass than the nearly complete sample. The high-M/L forecast finds MBAs as faint as $H \approx 12$ that should be measurable at S/N = 5 using only LSST data, and $H \approx 13$ with strong pre-LSST data (see Figure 5).

We also find that LSST can provide useful information on the masses of ≈ 20 Jupiter Trojans, if their tracers have good pre-LSST astrometry. This will be a substantial improvement over current knowledge—only three Trojan masses are known, possibly three more learned from the Lucy spacecraft over the coming 8 yr.

We have provided tables of the deflectors with prospects for mass detection, and the nature of the encounters that inform their masses: the tracer involved, geometry of the event, and forecasted accuracies on the impulse applied to each tracer.

Our forecasts incorporate a simplified model of the Yarkovsky effect as an azimuthal acceleration given by a time-independent A_2 value for each tracer MBA. In practice, it may be necessary to incorporate more complex radiation forces, but we believe the single parameter captures most of the resultant uncertainties in deflector mass. Another effect that could interfere with attaining the Fisher-forecasted

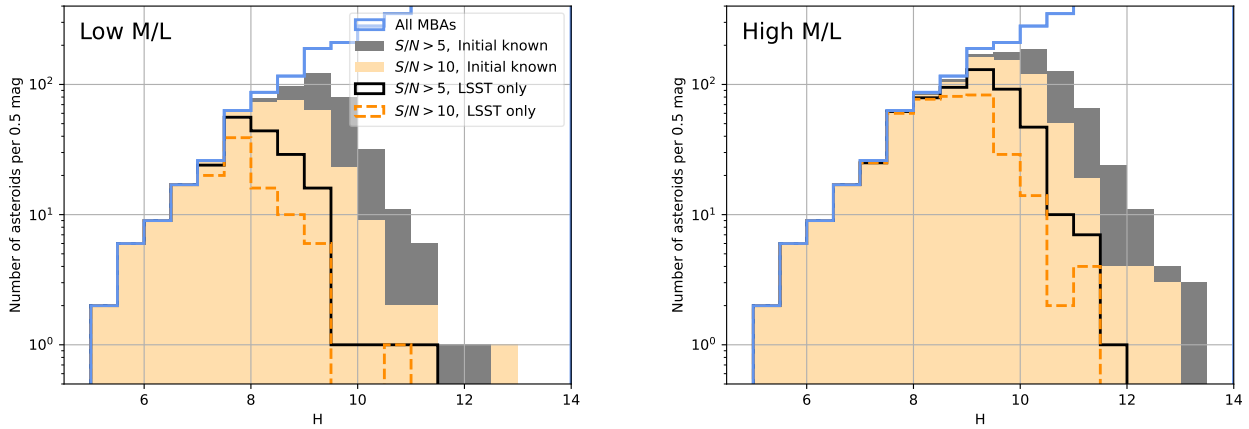


Figure 5. Histogram of the number of MBAs that are forecasted to attain S/N > 10 or S/N > 5 measurements of their masses by the use of LSST astrometry on tracer asteroids. The filled histograms assume that pre-LSST data establish the initial state vectors of the tracer MBAs. The open histograms show the conservative limit when only LSST observations are used. The left and right panels show results from assuming low-M/L and high-M/L asteroids, respectively. In each case, the mass census is nearly complete to some H value, and we forecast S/N > 5 mass measures for individual MBAs up to ≈4 mag fainter than the completion point enabled by unusually fortunate close encounters with known MBAs.

Table 1
Mutual Encounters with Impulse $>10^{-6}$ m s^{-1} and $b < 0.1$ au

Deflector (MPC Number)	Tracer (MPC Number)	MJD	Impact Vector b (10^{-3} au, ICRS)	Relative v (km s^{-1})	Impulse[a] (m s^{-1})	σ_I[b] (10^{-6} m s^{-1})		
00003	K05Ut7	61714.1	(+2.3565, +3.4339, −2.2074)	5.66	5.2×10^{-3}	422.8	24.2	19.5
00003	K15XC4	62408.0	(−5.7491, +1.2045, +0.3681)	4.22	5.6×10^{-3}	92.6	50.8	49.2
00003	K20T68	63961.9	(−10.9372, −2.1611, −4.6625)	1.80	6.4×10^{-3}	558.8	469.7	426.3
00003	k4081	63427.2	(−2.0037, +0.9809, −1.0470)	8.00	7.1×10^{-3}	25.1	11.2	8.8
00003	87536	62082.3	(−0.0713, +4.5060, +1.2259)	4.19	7.1×10^{-3}	9.4	5.8	4.7
…	…	…	…	…	…	…	…	…
T9937	K16Cf7	61499.9	(+0.0002, +0.0038, −0.0030)	2.09	1.9×10^{-6}	939.5	22.2	20.4
K10R84	K12U15	61374.7	(−0.0011, −0.0008, +0.0013)	4.36	1.0×10^{-6}	352.5	12.2	10.4
j8271	K10RM1	62171.1	(−0.0249, −0.1782, −0.0709)	0.00	1.7×10^{-5}	108.3	32.4	25.6
K10RM1	j8271	62171.1	(+0.0249, +0.1782, +0.0709)	0.00	2.4×10^{-6}	31.2	9.1	7.5

Notes.
[a] Impulse values assume the high-M/L value of $M_{10} = 10^{-13} M_\odot$ for all MBAs. Divide by 5 to obtain impulses if the low M/L ratio is chosen instead.
[b] The three columns give values in scenarios where only LSST data are used, where pre-LSST data determine the tracer asteroids' initial state vectors, and where the Yarkovsky acceleration A_2 is also known, respectively.

(This table is available in its entirety in machine-readable form in the online article.)

Table 2
Forecasted Mass Uncertainties for Deflector MBAs with S/N > 3

MPC Number	Designation	H	$\sigma_M/M_\odot$[a]			Nominal S/N[a]		
3	Juno	5.18	4.6×10^{-14}	1.7×10^{-14}	1.0×10^{-14}	1692.4	4701.1	7634.5
5	Astraea	6.99	7.7×10^{-15}	4.0×10^{-15}	3.4×10^{-15}	835.5	1596.8	1882.6
6	Hebe	5.61	5.0×10^{-14}	1.8×10^{-14}	1.1×10^{-14}	867.8	2413.9	3847.5
7	Iris	5.67	2.3×10^{-14}	5.2×10^{-15}	2.3×10^{-15}	1744.6	7731.8	17574.3
8	Flora	6.62	6.0×10^{-15}	1.7×10^{-15}	1.1×10^{-15}	1795.0	6369.0	9805.2
…	…	…	…	…	…	…	…	…
57276	2001 QP139	12.94	5.4×10^{-14}	5.1×10^{-16}	3.3×10^{-16}	0.0	3.4	5.2
66018	1998 QA31	13.37	7.3×10^{-14}	3.0×10^{-16}	2.4×10^{-16}	0.0	3.2	4.0
67138	2000 AF165	13.59	4.1×10^{-14}	2.3×10^{-16}	1.9×10^{-16}	0.0	3.1	3.6
76867	2000 YM5	10.48	2.4×10^{-13}	1.5×10^{-14}	1.4×10^{-14}	0.2	3.5	3.6

Note.
[a] The three columns give values in scenarios where only LSST data are used, where pre-LSST data determine the tracer asteroids' initial state vectors, and where the Yarkovsky acceleration A_2 is also known, respectively. All columns assume the "high-M/L" scenario in which an $H = 10$ MBA has mass $2 \times 10^{-14} M_\odot$. The low-M/L scenario reduces the S/N by a factor of 5.

(This table is available in its entirety in machine-readable form in the online article.)

uncertainties on deflector masses is that each tracer is subject to deflections by all of the millions of other MBAs, which are individually undetectable in its trajectory, but combine to cause a Brownian motion that would be a source of astrometric noise. In G. M. Bernstein (2025), we estimated this noise and found it would indeed be negligible for the present purposes.

On the other hand, our forecasts are conservative in that they do not include information that might be gained from the millions of MBAs that LSST will discover. While these will be fainter than the known MBAs—and hence have larger astrometric errors, sparse or absent pre-LSST measurements, and hence larger σ_M values—their shear numbers may lead to more very-low-b, low-v encounters that produce large, more easily measured impulses.

The forecasts show that there is potentially much to gain by incorporation of pre-LSST astrometry of the tracer MBAs into the inferences on the impulses that occur during LSST. Such data would need to be placed onto the Gaia reference frame and have reliable uncertainties estimated, if the original reported positions did not already include these. We have not delved into this exercise, but it would be essential to getting the most out of LSST. There is reason for optimism that the "Initial-known" scenario is closer to reality, because >90% of the deflectors that would attain S/N > 5 in the high-M/L, Initial-known scenario will deflect at least one tracer that has measurements in the Gaia Focused Product Release table of solar system object positions (Gaia Collaboration et al. 2023). Most of these have multiple Gaia targets among their 20 most-deflected tracers. Furthermore, the >20 yr span of Gaia+LSST observations opens the door to measuring impulses that occurred before Year 2 of LSST.

The computational burden of determining the hundreds of MBA masses that are detectable in LSST data should be small. We have seen that only a few thousand out of the many millions of LSST-detectable asteroids will need to have precision orbit fitting executed. G. M. Bernstein (2025) showed that it is only necessary to include the masses of a few hundred MBAs as free parameters in a global ephemeris fit in order to reduce errors from unmodeled masses to insignificant levels. This will be a relatively easy task.

The science gains from the harvest of LSST-based asteroid-mass estimates will be large, not only because the well-measured sample will be much larger and span a wider size/mass range. From the LSST (and precursor) data, we will be able to produce a homogenously reduced, well-characterized catalog, which can exploit nondetection data as well as S/N > 5 detections in order to characterize the mass-to-light relation $M(H)$ as a function of H, color, and dynamical state. One might, for example, expect the $M(H)$ relation to change slope if porosity is correlated with size; to vary with color or spectral class if these are indicative of bulk composition; and with location in the main belt, if inner and outer MBAs were formed in regions of the solar system with different local conditions.

It should also be possible to estimate the mean relation of mass to H for classes of MBAs divided into orbital and/or photometric classes, even when individual objects' masses have insufficient S/N for detection. This could be done by summing ("stacking") the information from all of their collective gravitational encounters with other MBAs. This would allow for characterization of the MBA physical properties across color and dynamical space for smaller bodies than those cataloged in this work.

Another scientific gain will be higher-precision solar system ephemerides. As explored in G. M. Bernstein (2025), producing mass estimates at S/N $\gtrsim$ 1 for $O(1000)$ asteroids, as we forecast for LSST, will greatly reduce the level of uncertainty in solar system ephemerides by improving the predictions of gravitational forces from the main belt.

In summary, we find that LSST should boost the number of MBAs with usefully measured masses (S/N $\gtrsim$ 5) from the tens into the hundreds, including nearly complete coverage of bodies with $H \lesssim 9$. With such studies, time is on the observer's side: the number of measurable encounters increases with time, the size of the perturbations from a given impulse grows with time, the ability to distinguish an impulse from the Yarkovsky effect improves with time, and the continued observations improve the sensitivity of the data.

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